

1(a) ANOVA: Infection Risk by Geographic Region

The GLM Procedure

Class Level Information			
Class	Levels	Values	
region	4	1	2 3 4

Number of Observations Read	113
Number of Observations Used	113

1(a) ANOVA: Infection Risk by Geographic Region

The GLM Procedure

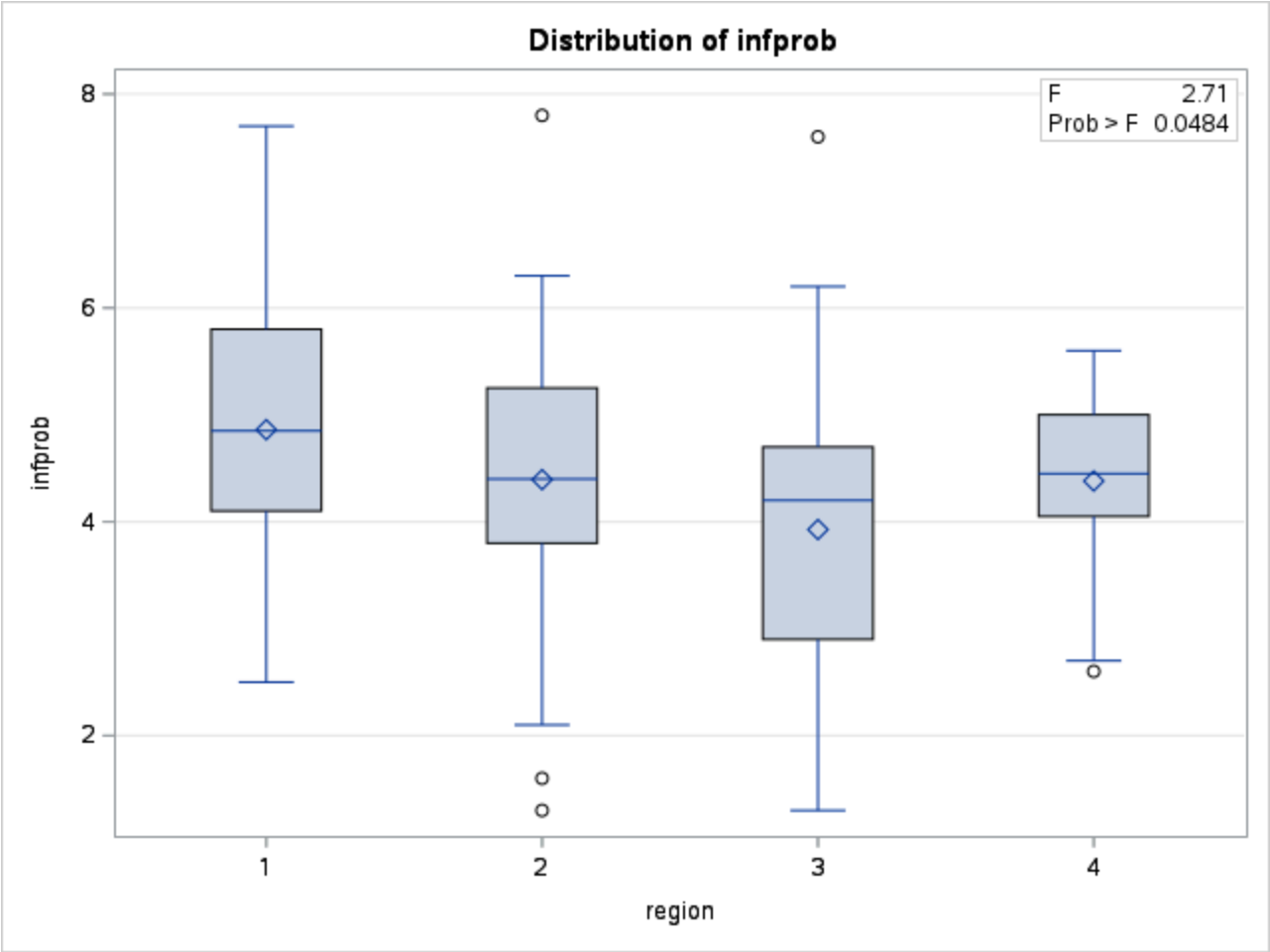
Dependent Variable: infprob

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	13.9969393	4.6656464	2.71	0.0484
Error	109	187.3828837	1.7191090		
Corrected Total	112	201.3798230			

R-Square	Coeff Var	Root MSE	infprob Mean
0.069505	30.10765	1.311148	4.354867

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	13.99693932	4.66564644	2.71	0.0484

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	13.99693932	4.66564644	2.71	0.0484



1(b) Tukey Pairwise Comparisons (90% Confidence)

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

1(b) Tukey Pairwise Comparisons (90% Confidence)

The GLM Procedure

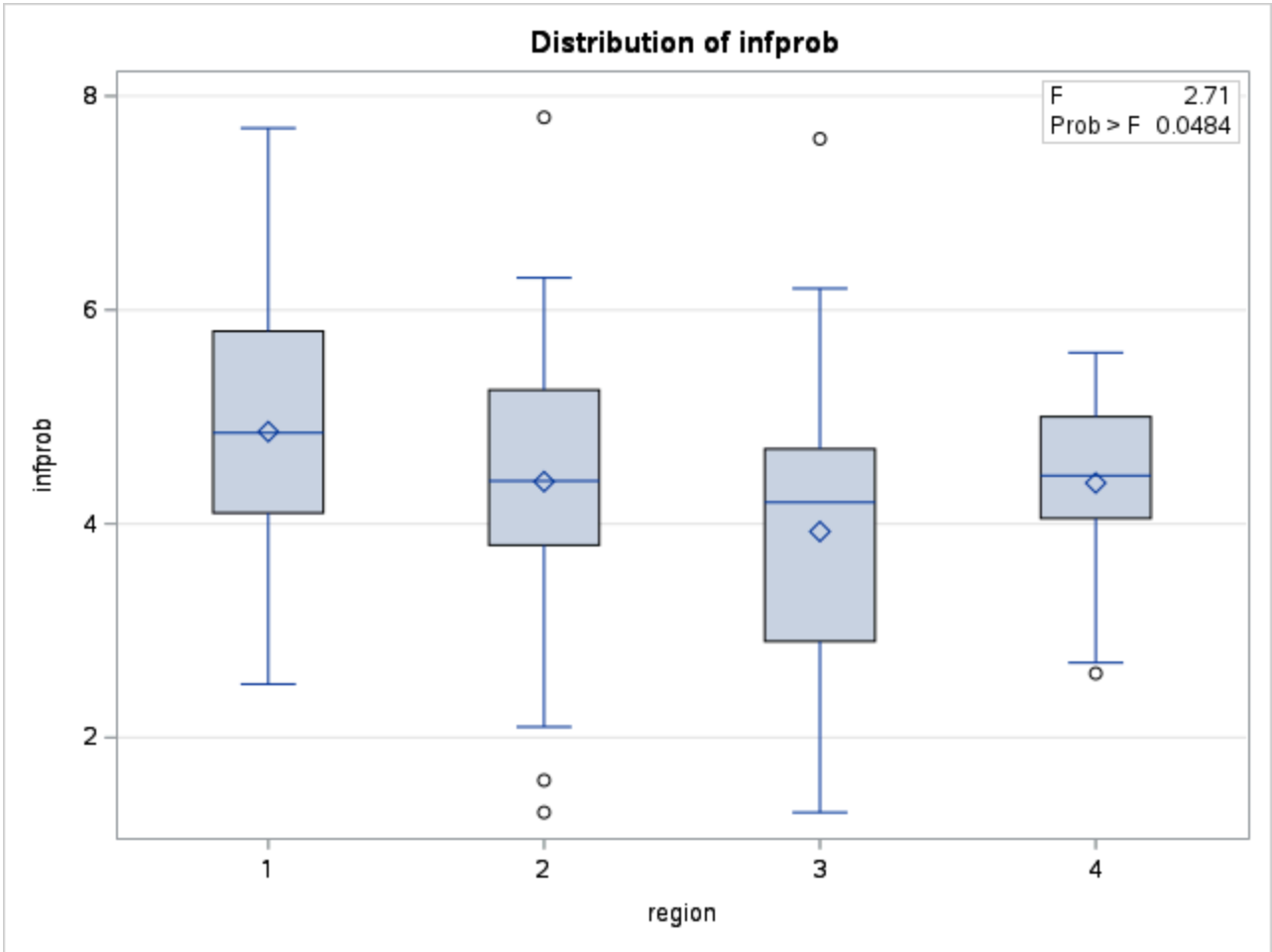
Dependent Variable: infprob

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	13.9969393	4.6656464	2.71	0.0484
Error	109	187.3828837	1.7191090		
Corrected Total	112	201.3798230			

R-Square	Coeff Var	Root MSE	infprob Mean
0.069505	30.10765	1.311148	4.354867

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	13.99693932	4.66564644	2.71	0.0484

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	13.99693932	4.66564644	2.71	0.0484



1(b) Tukey Pairwise Comparisons (90% Confidence)

The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

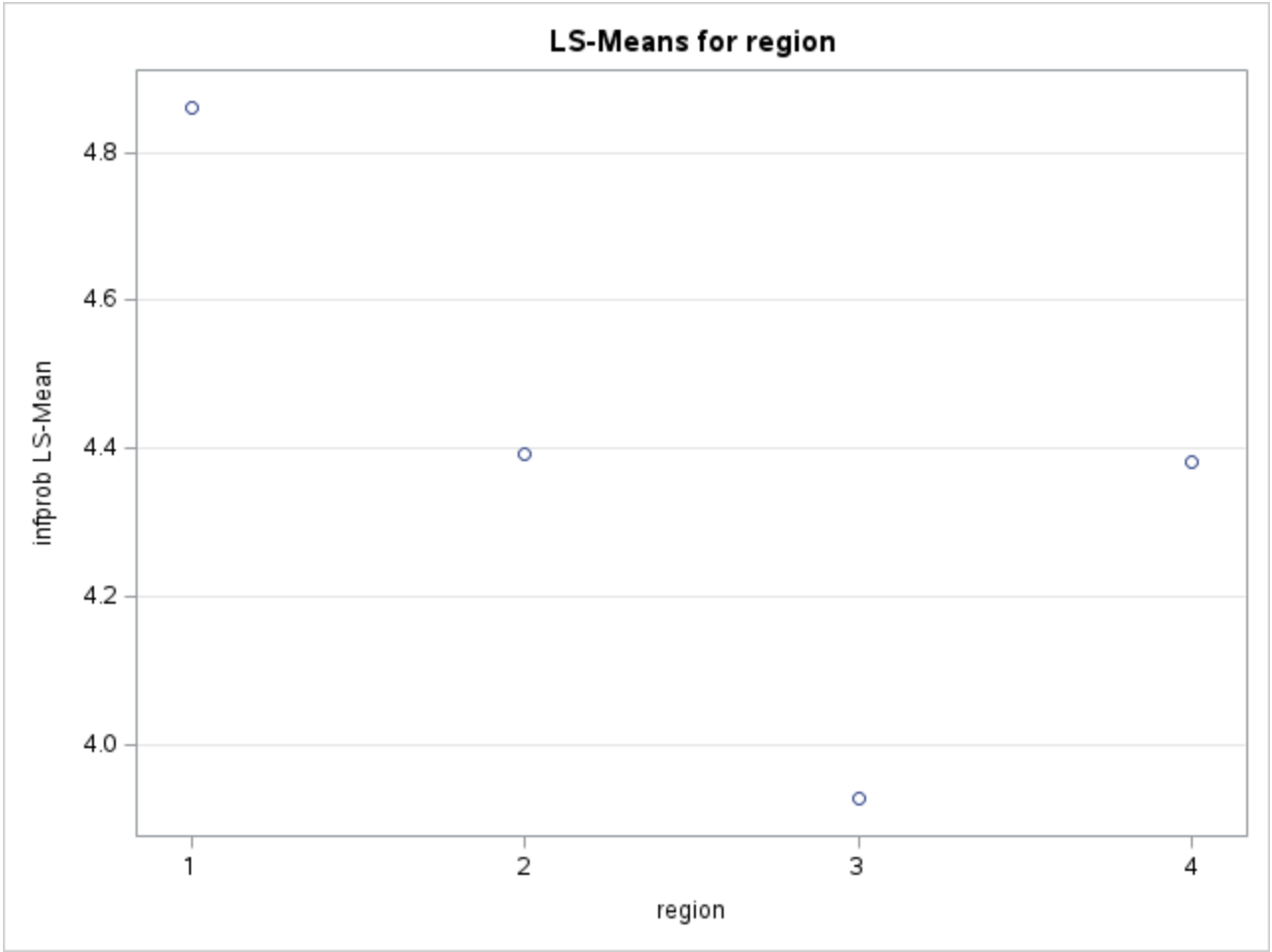
region	infprob LSMEAN	LSMEAN Number
1	4.86071429	1
2	4.39375000	2
3	3.92702703	3
4	4.38125000	4

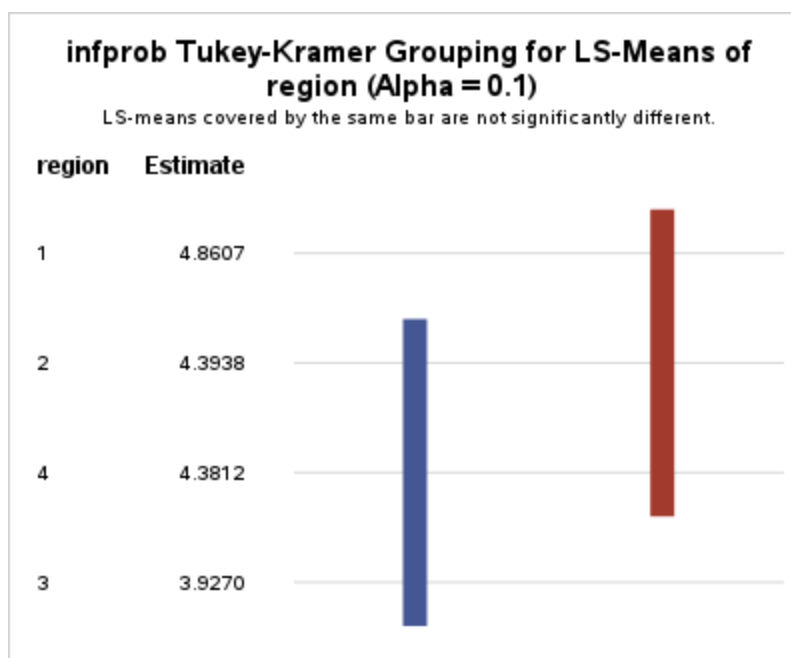
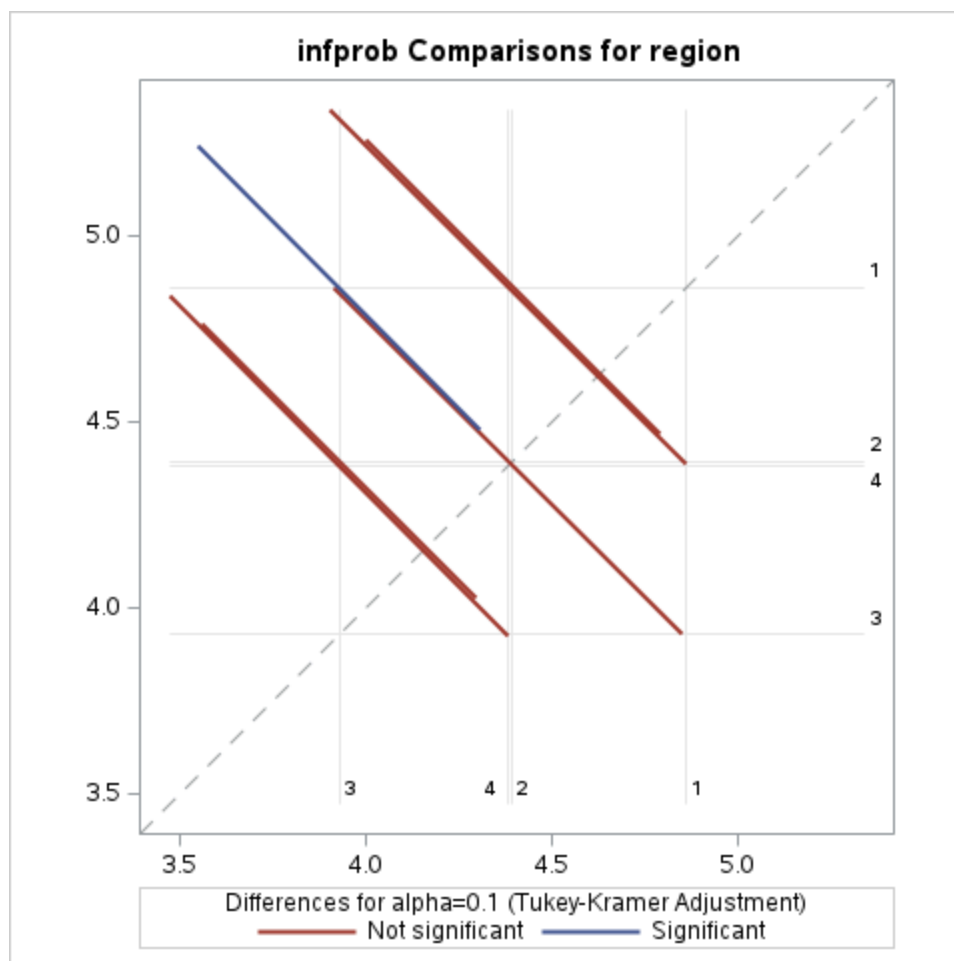
Least Squares Means for effect region Pr > t for H0: LSMean(i)=LSMean(j)				
Dependent Variable: infprob				
i/j	1	2	3	4
1		0.5169	0.0270	0.6490
2	0.5169		0.4563	1.0000
3	0.0270	0.4563		0.6546

Least Squares Means for effect region				
Pr > t for H0: LSMean(i)=LSMean(j)				
Dependent Variable: infprob				
i/j	1	2	3	4
4	0.6490	1.0000	0.6546	

region	infprob LSMEAN	90% Confidence Limits	
1	4.860714	4.449653	5.271776
2	4.393750	4.009237	4.778263
3	3.927027	3.569437	4.284617
4	4.381250	3.837467	4.925033

Least Squares Means for Effect region				
i	j	Difference Between Means	Simultaneous 90% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	0.466964	-0.319842	1.253770
1	3	0.933687	0.172095	1.695280
1	4	0.479464	-0.473405	1.432333
2	3	0.466723	-0.267275	1.200721
2	4	0.012500	-0.918461	0.943461
3	4	-0.454223	-1.363975	0.455529





1(c) Bonferroni Pairwise Comparisons (90% Confidence)

The GLM Procedure

Class Level Information		
Class	Levels	Values

Class Level Information		
Class	Levels	Values
region	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

1(c) Bonferroni Pairwise Comparisons (90% Confidence)

The GLM Procedure

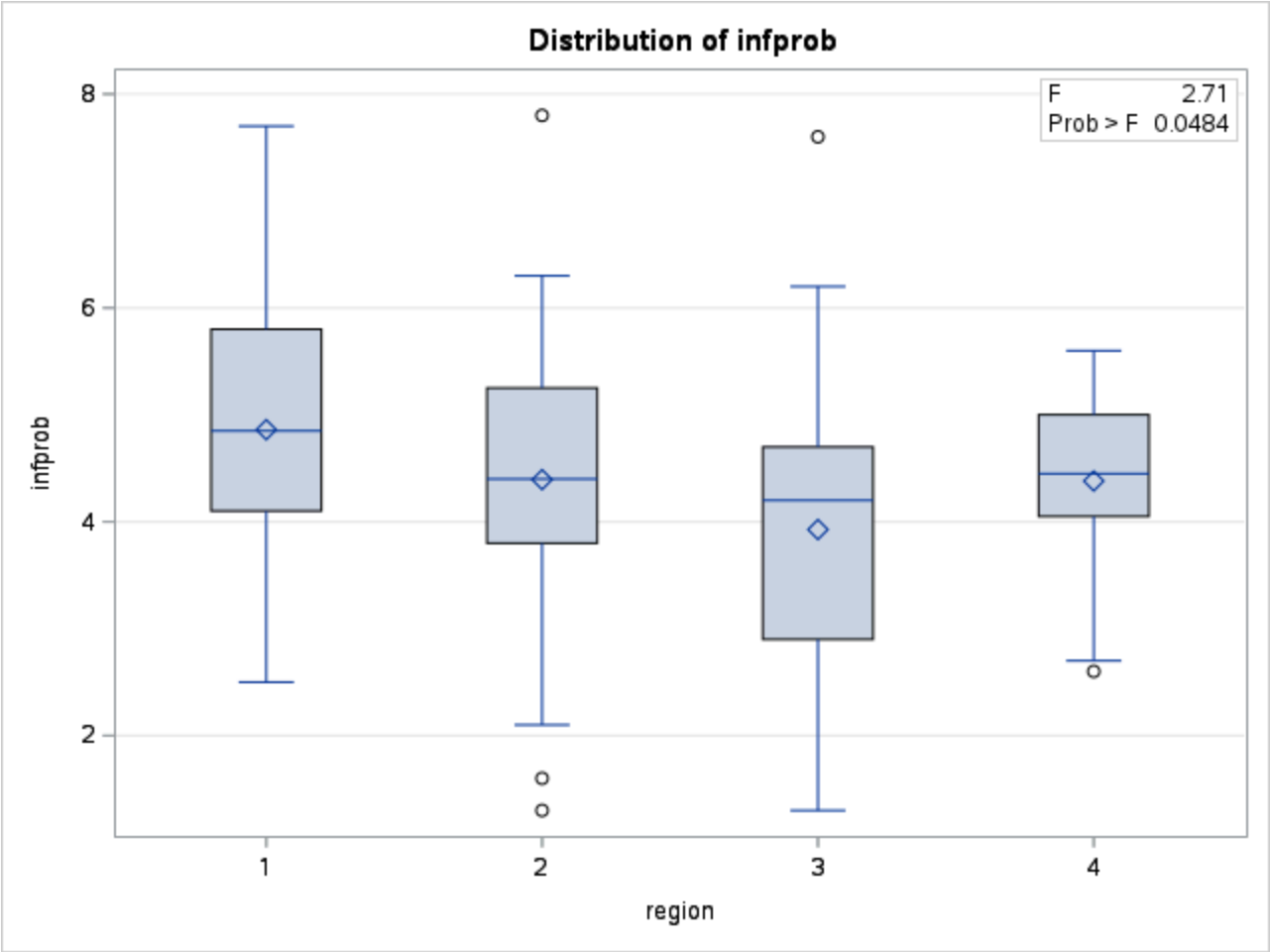
Dependent Variable: infprob

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	13.9969393	4.6656464	2.71	0.0484
Error	109	187.3828837	1.7191090		
Corrected Total	112	201.3798230			

R-Square	Coeff Var	Root MSE	infprob Mean
0.069505	30.10765	1.311148	4.354867

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	13.99693932	4.66564644	2.71	0.0484

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	13.99693932	4.66564644	2.71	0.0484



1(c) Bonferroni Pairwise Comparisons (90% Confidence)

The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Bonferroni

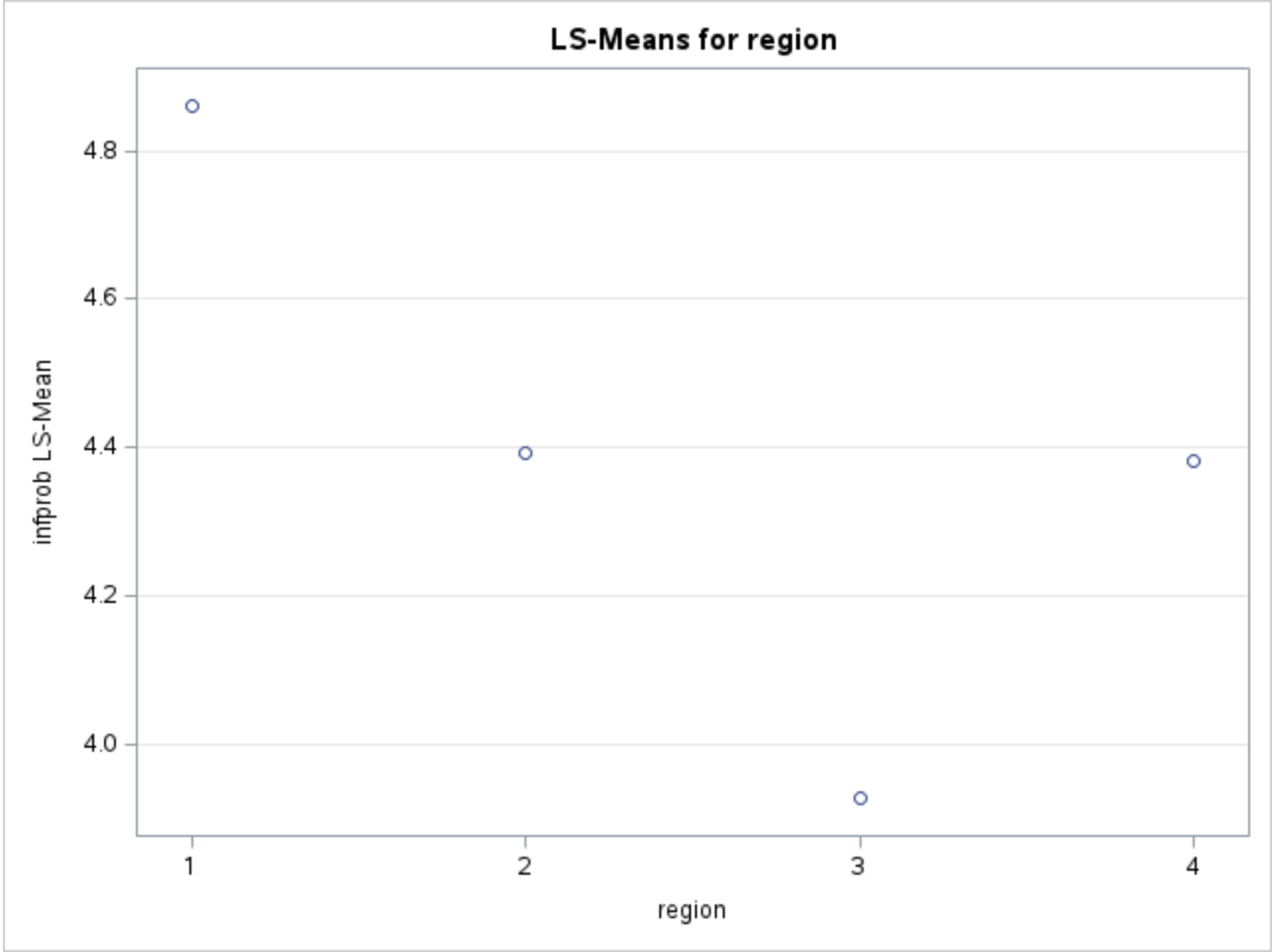
region	infprob LSMEAN	LSMEAN Number
1	4.86071429	1
2	4.39375000	2
3	3.92702703	3
4	4.38125000	4

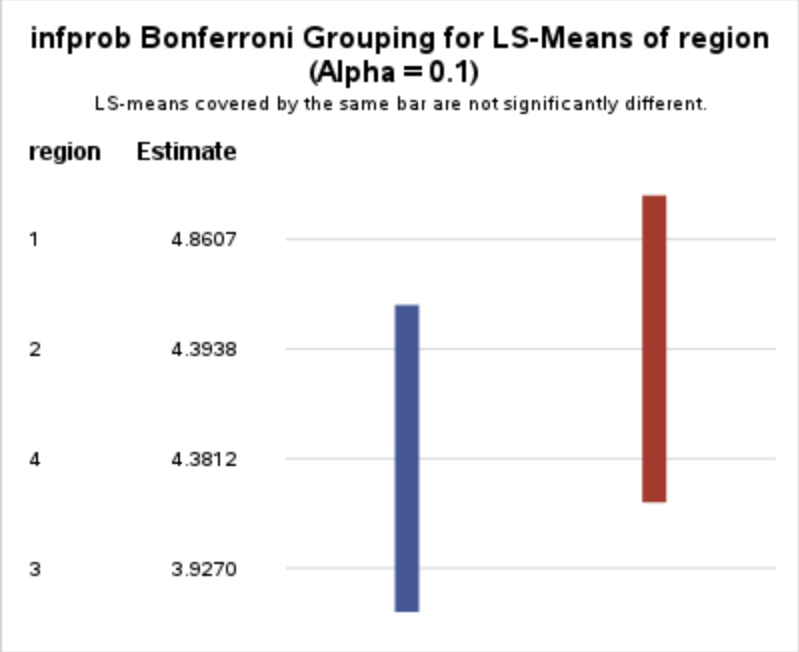
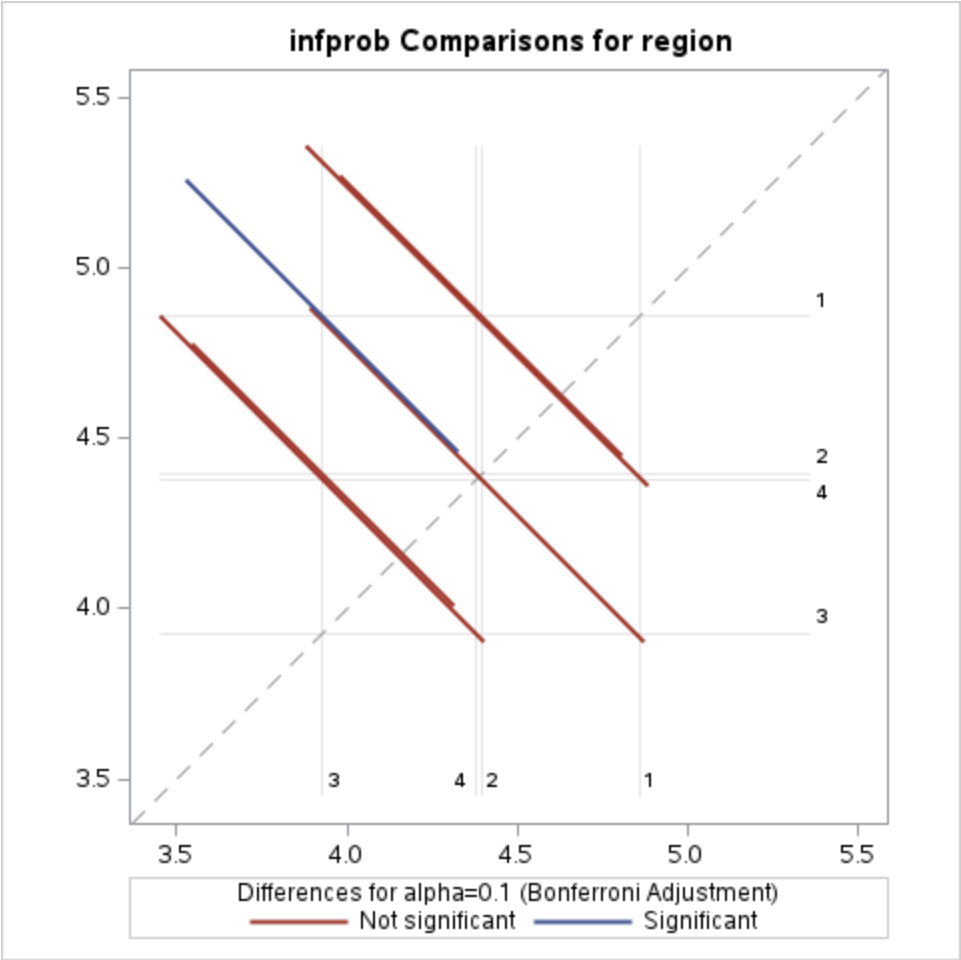
Least Squares Means for effect region Pr > t for H0: LSMean(i)=LSMean(j)				
Dependent Variable: infprob				
i/j	1	2	3	4
1		1.0000	0.0320	1.0000
2	1.0000		0.8593	1.0000
3	0.0320	0.8593		1.0000
4	1.0000	1.0000	1.0000	

region	infprob LSMEAN	90% Confidence Limits	
1	4.860714	4.449653	5.271776
2	4.393750	4.009237	4.778263
3	3.927027	3.569437	4.284617

region	infprob LSMEAN	90% Confidence Limits	
4	4.381250	3.837467	4.925033

Least Squares Means for Effect region				
i	j	Difference Between Means	Simultaneous 90% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	0.466964	-0.358020	1.291948
1	3	0.933687	0.135140	1.732235
1	4	0.479464	-0.519641	1.478570
2	3	0.466723	-0.302890	1.236336
2	4	0.012500	-0.963634	0.988634
3	4	-0.454223	-1.408119	0.499673





Question 2: ANOVA of Infection Risk by Age Group

The GLM Procedure

Class Level Information		
Class	Levels	Values

Class Level Information		
Class	Levels	Values
AgeGroup	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

Question 2: ANOVA of Infection Risk by Age Group

The GLM Procedure

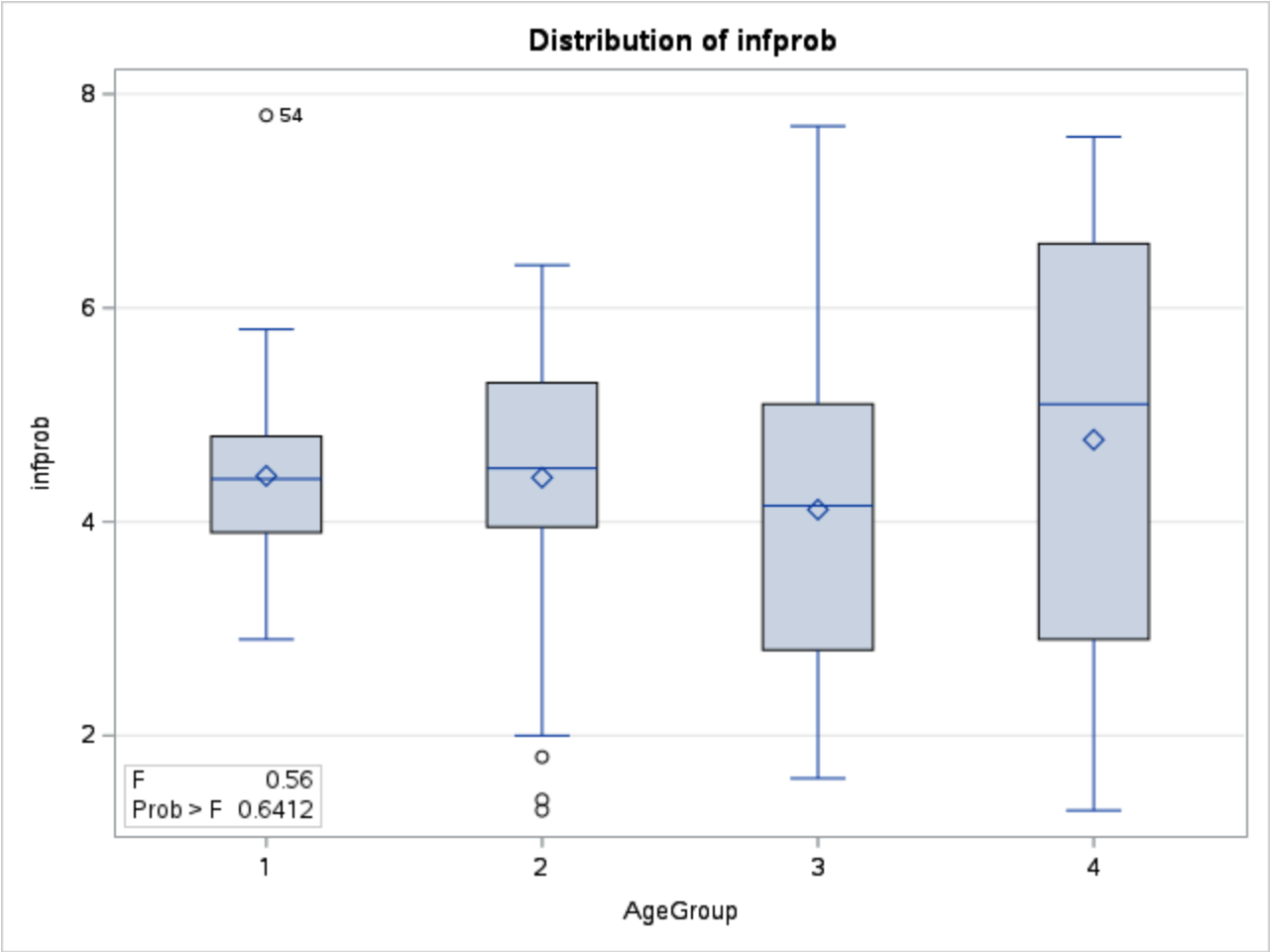
Dependent Variable: infprob

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3.0677159	1.0225720	0.56	0.6412
Error	109	198.3121071	1.8193771		
Corrected Total	112	201.3798230			

R-Square	Coeff Var	Root MSE	infprob Mean
0.015233	30.97323	1.348843	4.354867

Source	DF	Type I SS	Mean Square	F Value	Pr > F
AgeGroup	3	3.06771587	1.02257196	0.56	0.6412

Source	DF	Type III SS	Mean Square	F Value	Pr > F
AgeGroup	3	3.06771587	1.02257196	0.56	0.6412



Question 2: ANOVA of Infection Risk by Age Group

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

Question 2: ANOVA of Infection Risk by Age Group

The GLM Procedure

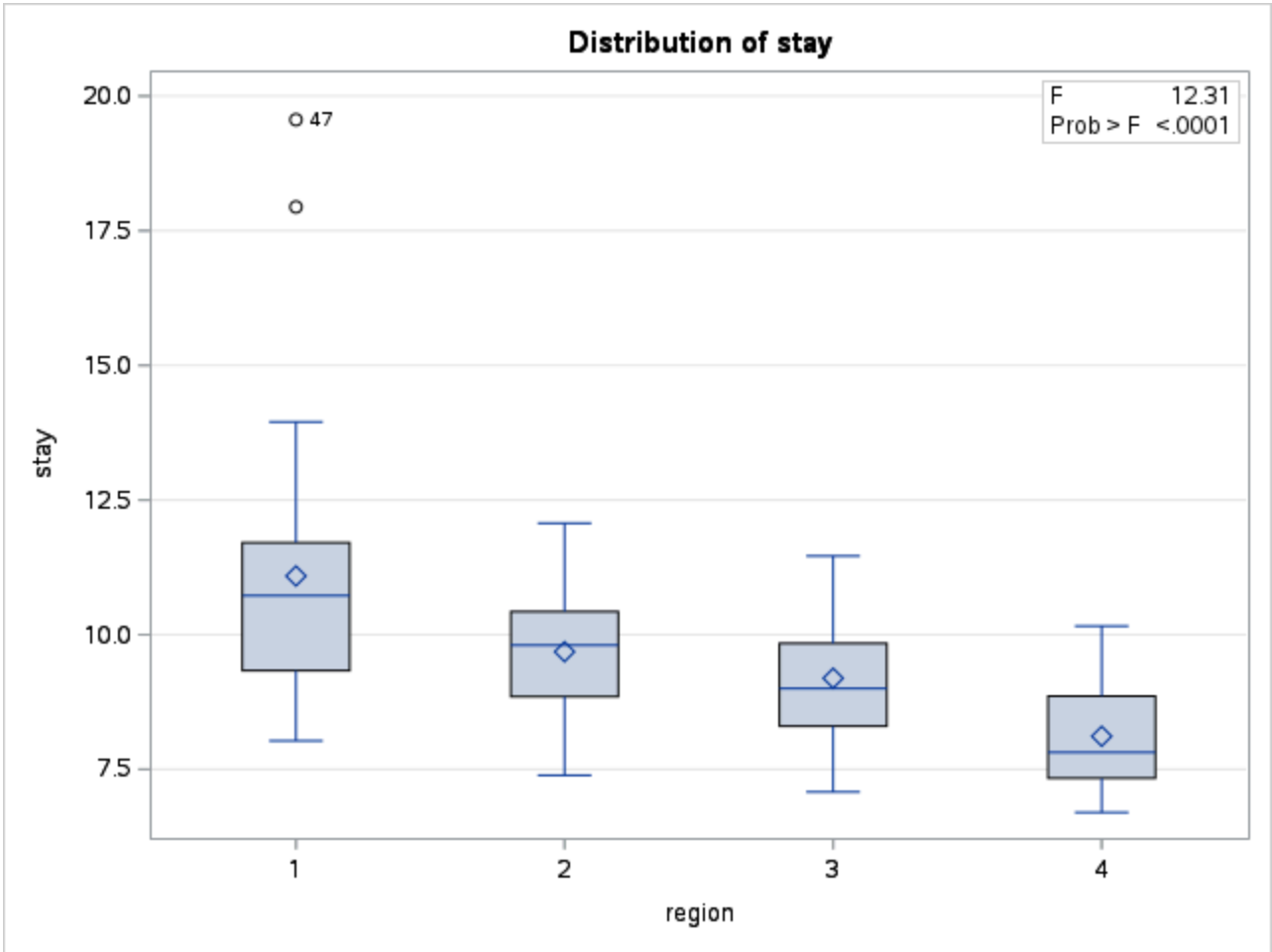
Dependent Variable: stay

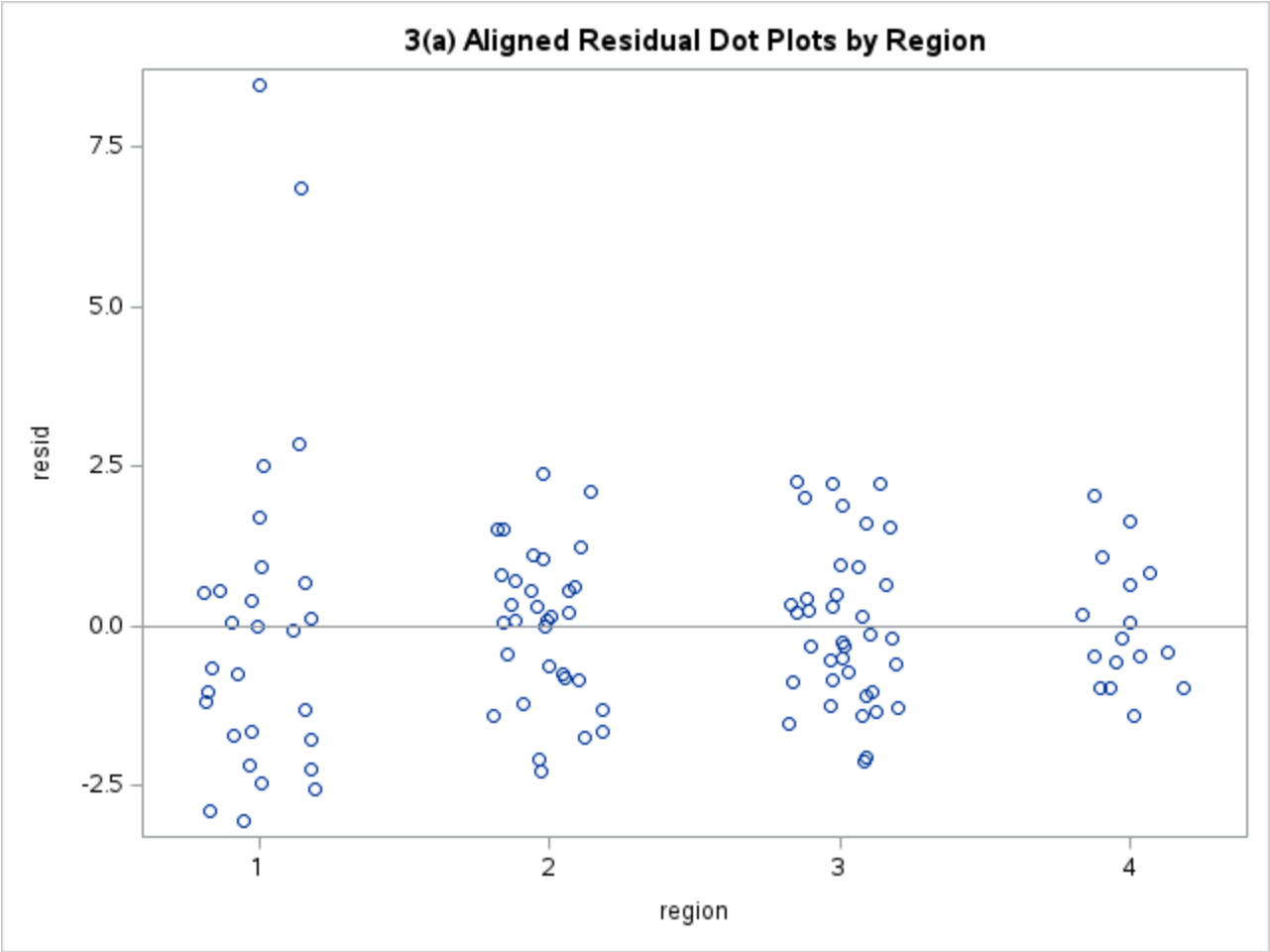
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	103.5541834	34.5180611	12.31	<.0001
Error	109	305.6561972	2.8041853		
Corrected Total	112	409.2103805			

R-Square	Coeff Var	Root MSE	stay Mean
0.253059	17.35608	1.674570	9.648319

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	103.5541834	34.5180611	12.31	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	103.5541834	34.5180611	12.31	<.0001





3(b) Brown-Forsythe Test (Original Length of Stay)

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

3(b) Brown-Forsythe Test (Original Length of Stay)

The GLM Procedure

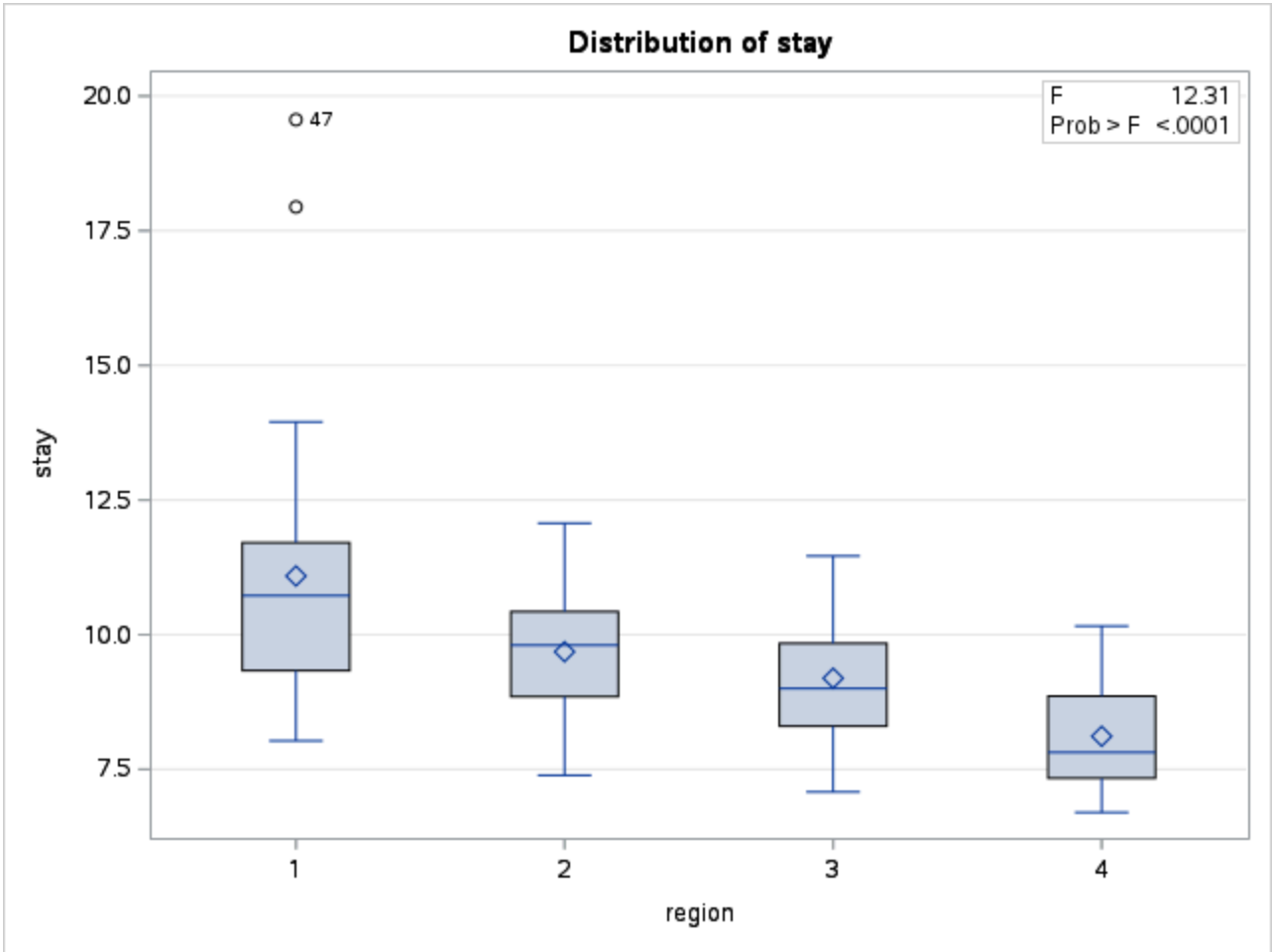
Dependent Variable: stay

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	103.5541834	34.5180611	12.31	<.0001
Error	109	305.6561972	2.8041853		
Corrected Total	112	409.2103805			

R-Square	Coeff Var	Root MSE	stay Mean
0.253059	17.35608	1.674570	9.648319

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	103.5541834	34.5180611	12.31	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	103.5541834	34.5180611	12.31	<.0001



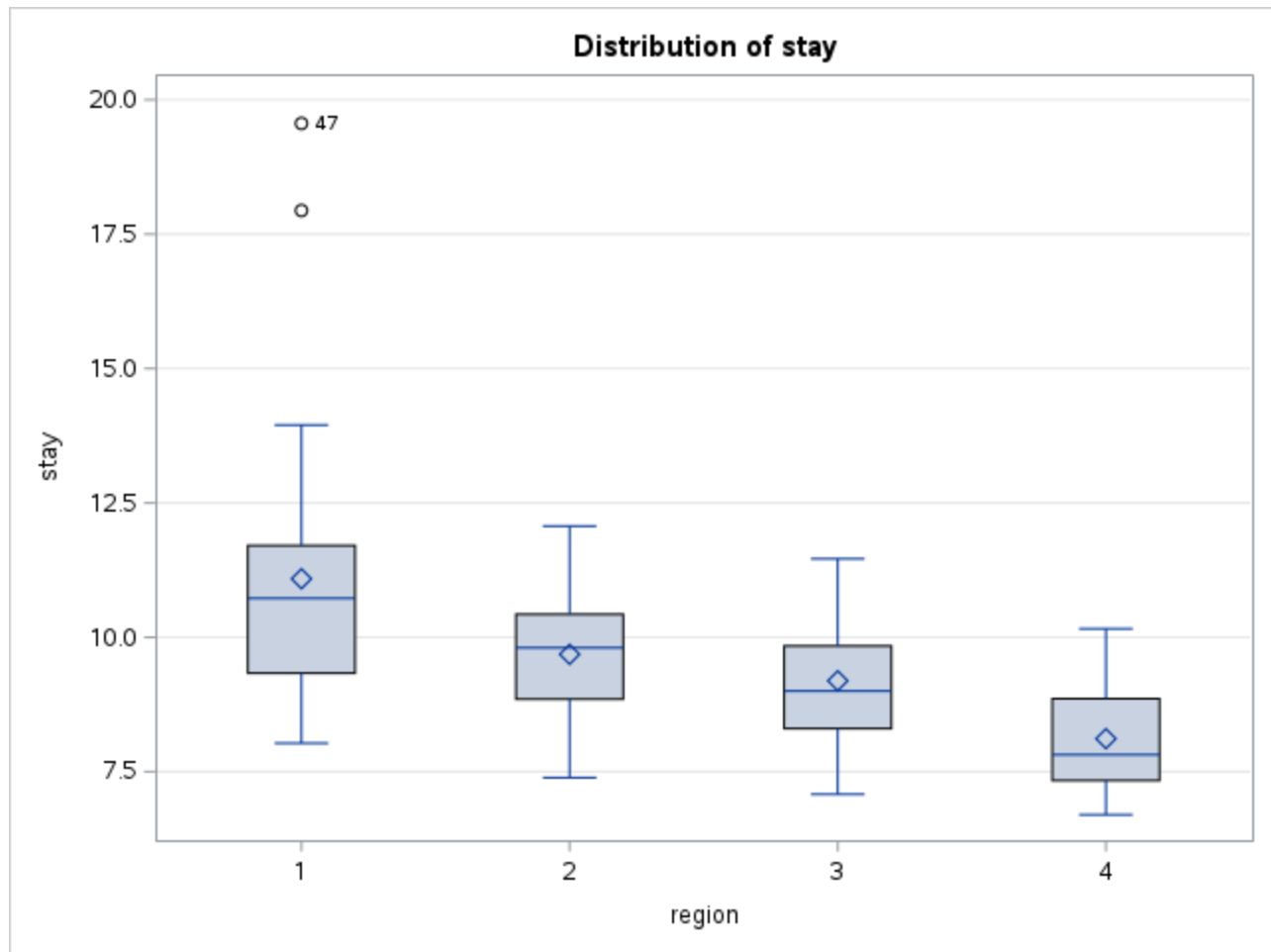
3(b) Brown-Forsythe Test (Original Length of Stay)

The GLM Procedure

Brown and Forsythe's Test for Homogeneity of stay Variance ANOVA of Absolute Deviations from Group Medians					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
region	3	17.2864	5.7621	4.33	0.0064
Error	109	145.2	1.3319		

3(b) Brown-Forsythe Test (Original Length of Stay)

The GLM Procedure



Level of region	N	stay	
		Mean	Std Dev
1	28	11.0889286	2.66961551
2	32	9.6834375	1.19293785
3	37	9.1913514	1.22498790
4	16	8.1137500	1.00312096

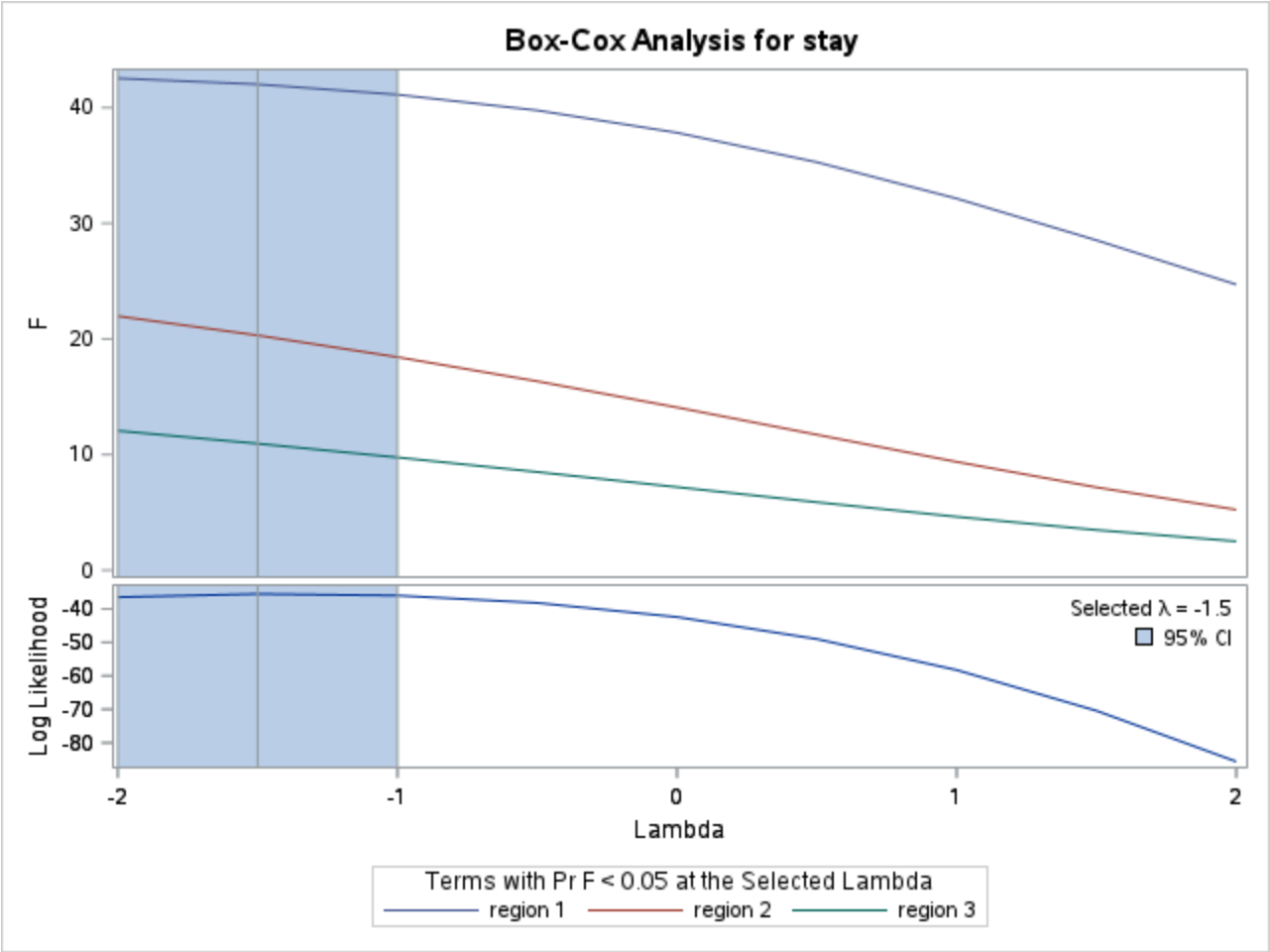
3(c) Mean and Std Dev of Length of Stay by Region

The MEANS Procedure

Analysis Variable : stay			
region	N Obs	Mean	Std Dev
1	28	11.0889286	2.6696155
2	32	9.6834375	1.1929378
3	37	9.1913514	1.2249879
4	16	8.1137500	1.0031210

3(d) Box-Cox Transformation Analysis

The TRANSREG Procedure



Dependent Variable BoxCox(stay)

Class Level Information			
Class	Levels	Values	
region	4	1	2 3 4

Number of Observations Read	113
Number of Observations Used	113

The TRANSREG Procedure Hypothesis Tests for BoxCox(stay)

Univariate ANOVA Table Based on the Usual Degrees of Freedom					
Source	DF	Sum of Squares	Mean Square	F Value	Liberal p
Model	3	0.001098	0.000366	15.01	>= <.0001
Error	109	0.002657	0.000024		
Corrected Total	112	0.003755			
The above statistics are not adjusted for the fact that the dependent variable was transformed and so are generally liberal.					

Root MSE	0.00494	R-Square	0.2923
Dependent Mean	0.64310	Adj R-Sq	0.2729
Coeff Var	0.76774	Lambda	-1.5000

3(e) ANOVA on Transformed Data ($Y' = 1 / \text{stay}$)

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

3(e) ANOVA on Transformed Data ($Y' = 1 / \text{stay}$)

The GLM Procedure

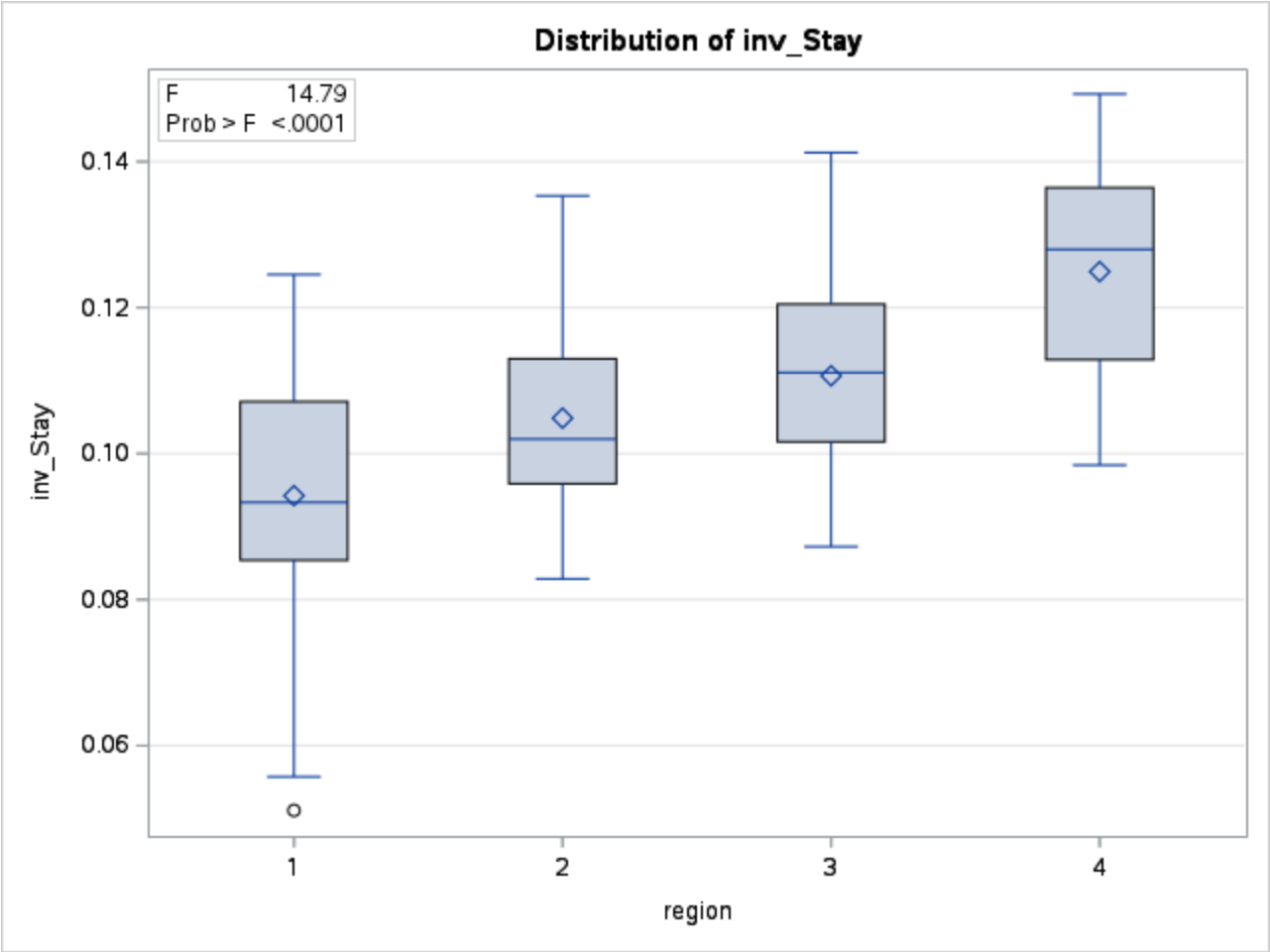
Dependent Variable: inv_Stay

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	0.01034953	0.00344984	14.79	<.0001
Error	109	0.02542843	0.00023329		
Corrected Total	112	0.03577796			

R-Square	Coeff Var	Root MSE	inv_Stay Mean
0.289271	14.27848	0.015274	0.106971

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	0.01034953	0.00344984	14.79	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	0.01034953	0.00344984	14.79	<.0001



3(f) Brown-Forsythe Test on Transformed Data (Alpha = 0.01)

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	4	1 2 3 4

Number of Observations Read	113
Number of Observations Used	113

3(f) Brown-Forsythe Test on Transformed Data (Alpha = 0.01)

The GLM Procedure

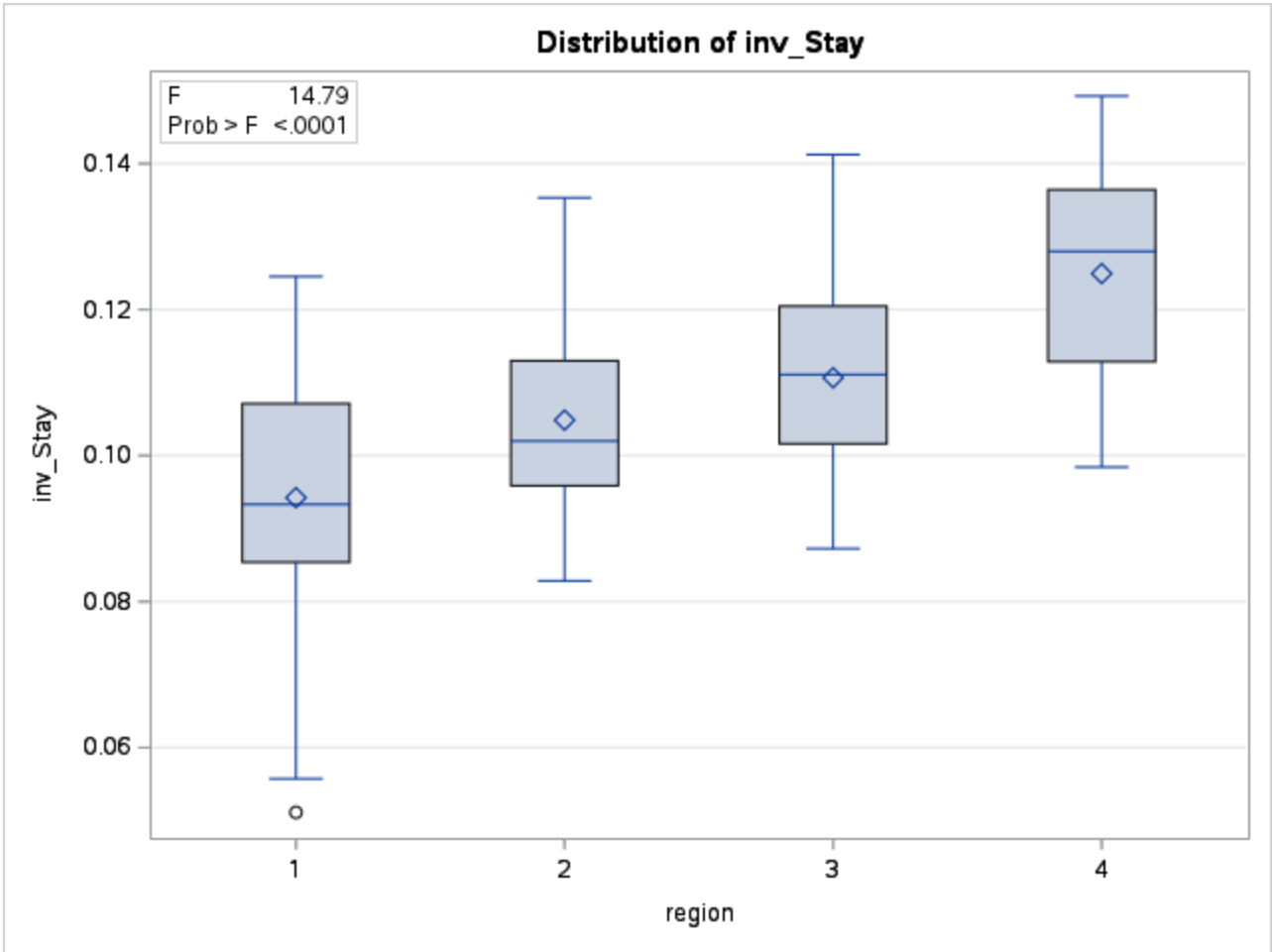
Dependent Variable: inv_Stay

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	0.01034953	0.00344984	14.79	<.0001
Error	109	0.02542843	0.00023329		
Corrected Total	112	0.03577796			

R-Square	Coeff Var	Root MSE	inv_Stay Mean
0.289271	14.27848	0.015274	0.106971

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	3	0.01034953	0.00344984	14.79	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	0.01034953	0.00344984	14.79	<.0001



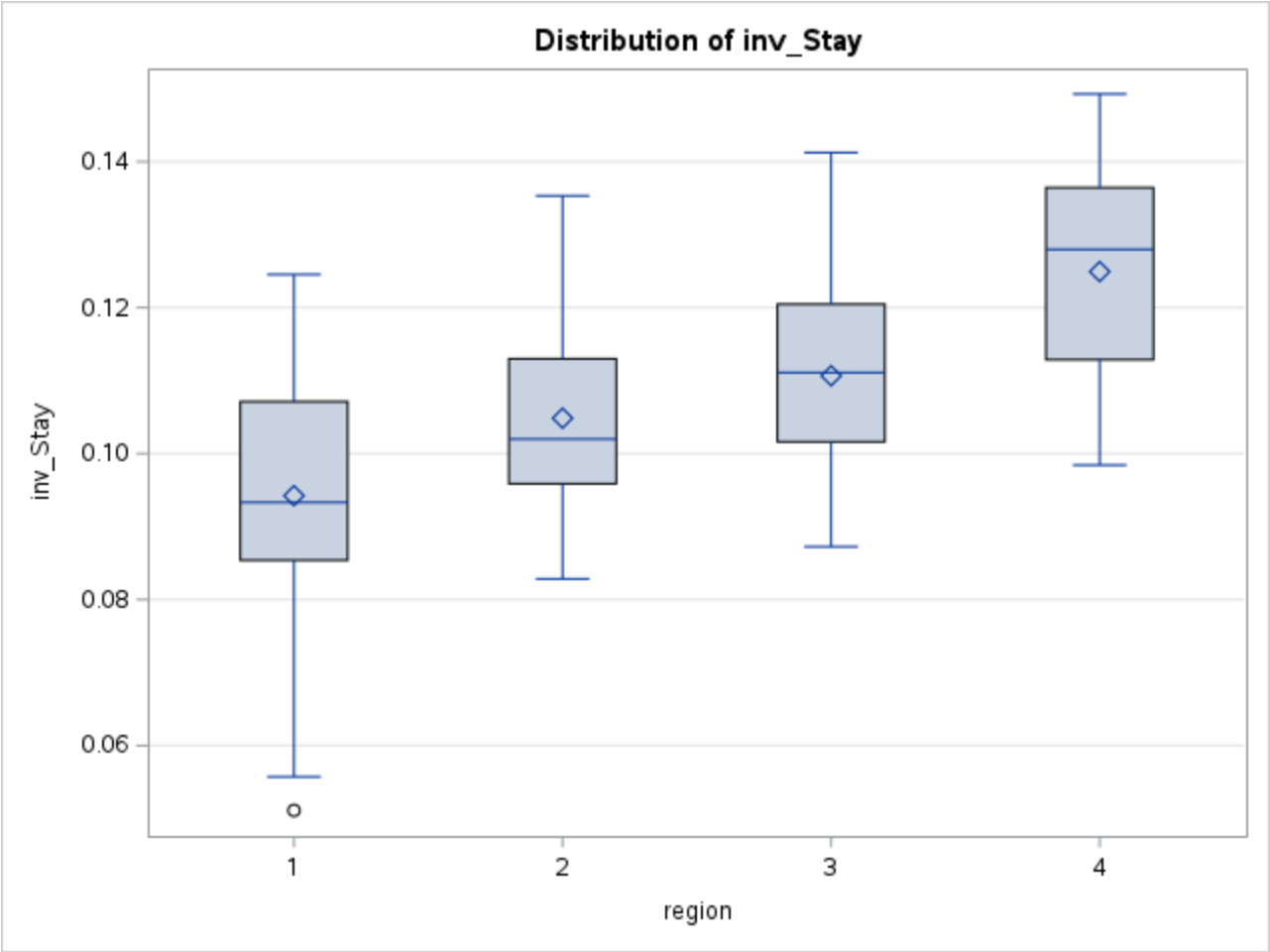
3(f) Brown-Forsythe Test on Transformed Data (Alpha = 0.01)

The GLM Procedure

Brown and Forsythe's Test for Homogeneity of inv_Stay Variance ANOVA of Absolute Deviations from Group Medians					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
region	3	0.000246	0.000082	0.97	0.4100
Error	109	0.00924	0.000085		

3(f) Brown-Forsythe Test on Transformed Data (Alpha = 0.01)

The GLM Procedure



Level of region	N	inv_Stay	
		Mean	Std Dev
1	28	0.09423962	0.01824992
2	32	0.10485885	0.01345123
3	37	0.11066240	0.01452734
4	16	0.12493647	0.01467252